

Branch Specialization Checkpoint

Weak Token-Router Supervision Sweep

Branch Specialization Project

2026-05-22

Checkpoint reason: gate compliance appears before causal branch modularity.

Previous checkpoint:

- weak token-router supervision at weight 0.05 produced near-oracle branch-level functional modularity;
- unconstrained learned token routing did not.

This checkpoint asks:

How much routing supervision is needed before learned gates become causal branch modularity, not just gate preference?

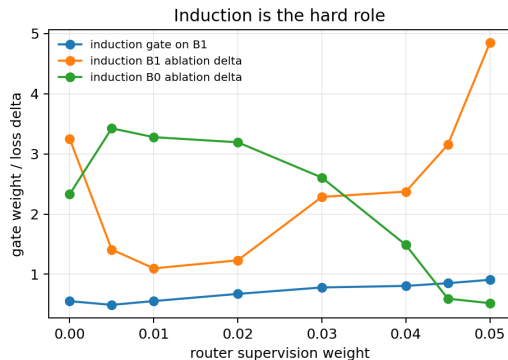
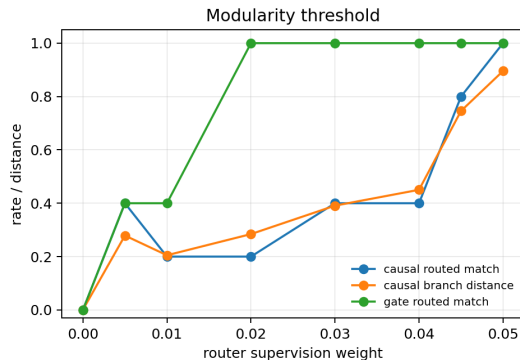
Same local-vs-induction competition task and branch model:

- config: `weak_token_router`;
- local weight 0.25, induction weight 1.0;
- five seeds per weight, 1200 training steps;
- branch modularity measured by branch ablation, not gate values alone.

Swept scored-position routing loss weights:

0.005, 0.01, 0.02, 0.03, 0.04, 0.045, 0.05

Main Evidence



Gates point the right way before causal ablations become modular. Reliable 5/5 causal modularity appears only at weight 0.05 in this sweep.

Threshold Table

Weight	Local acc.	Ind. acc.	Gate match	Causal match	Branch dist.
0.000	1.0000	0.9989	0.00	0.00	0.0006
0.005	1.0000	0.9978	0.40	0.40	0.2789
0.010	0.9998	0.9987	0.40	0.20	0.2050
0.020	0.9994	0.9985	1.00	0.20	0.2844
0.030	0.9999	0.9994	1.00	0.40	0.3910
0.040	1.0000	0.9983	1.00	0.40	0.4504
0.045	0.9999	0.9985	1.00	0.80	0.7459
0.050	0.9998	0.9983	1.00	1.00	0.8957

Causal match means local top branch is 0 and induction top branch is 1 under branch ablation.

Per-Seed Pattern

Weight	Local top branches	Induction top branches
0.005	{0:5}	{0:3, 1:2}
0.010	{0:5}	{0:4, 1:1}
0.020	{0:5}	{0:4, 1:1}
0.030	{0:5}	{0:3, 1:2}
0.040	{0:5}	{0:3, 1:2}
0.045	{0:5}	{0:1, 1:4}
0.050	{0:5}	{1:5}

Local becomes branch-0-specific at every nonzero weight. Induction is the limiting role and flips reliably only near the top of the sweep.

At weight 0.02:

- gate routed match = 1.00;
- causal routed match = 0.20.

At weight 0.05:

- gate routed match = 1.00;
- causal routed match = 1.00.

The router can assign positions to the intended branches before the branch computations themselves become role-separated.

Supported in this toy setup:

- A routing-pressure threshold exists for causal modularity.
- Gate metrics alone can overstate functional modularity.
- Causal branch ablation is necessary for the modularity claim.

Current threshold estimate:

5/5 causal modularity appears at 0.05; 0.045 is near-threshold at 4/5; weights at or below 0.04 are mixed.

Stop adding labeled-router weights for now.

Next decisive experiment:

Test unlabeled routing pressure, such as entropy or load-balancing regularization, and ask whether it can produce any similar causal modularity shift without role labels.

This distinguishes labeled routing sufficiency from spontaneous or regularization-induced modularity.

Sources and Provenance

- Report: `doc/phase3_toy_weak_router_sweep.md`
- Model script: `scripts/toy_branch_isolation_intervention.py`
- Analysis script: `scripts/analyze_weak_router_sweep.py`
- Sweep output: `results/phase3_toy_weak_token_router_sweep_analysis/`
- Deck data snapshot: `presentations/2026-05-22-1354-weak-router-sweep/data/`